



Maintenance Schedule

Diesel engine

12V4000G73 / T1238A34-2150

16V4000G73 / T1638A34-2152

Application group 3B

MS50010/04E

12V4000G73
16V4000G73

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL 1 to QL4.

QL 1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

1.3 Load profile

Load profile

Load	110%	100%	90%	80%	30%
Corresponding operating time	1%	5%	3%	78%	13%

1.4 Maintenance schedule matrix

0 to 10,500 Operating hours

Item	Limit	Operating hours [h]																							
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500		
Engine																									
Emergency air shut-off flaps	1 w																								
Engine operation	-	X																							
Centrifugal oil filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Automatic engine oil filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Engine oil filter	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Emergency air shut-off flaps	3 m			X		X		X		X		X		X		X		X		X		X			
Engine mounts	1 a			X		X		X		X		X		X		X		X		X		X			
Fuel filter	2 a			X		X		X		X		X		X		X		X		X		X			
Belt drive	2 a							X						X						X					
Air filters	3 a							X						X						X					
Valve gear	18 a			X				X						X						X					
Cylinder heads	18 a							X						X						X					
Fuel injectors	18 a													X											
Engine governor	18 a													X											
Combustion chambers	18 a													X											
Crankcase breathers	18 a																								
Engine mounts	5 a																								
Rubber sleeves	6 a																								
Component maintenance	18 a																								
Turbochargers	18 a																								
HP fuel pump	18 a																								
Hose lines	6 a																								
Cylinder heads	18 a																								
Extended component maintenance	18 a																								
Engine mounts	18 a																								
Torsional resilient coupling	18 a																								
Automatic engine oil filter	18 a																								
w = weeks m = months a = years																									

11,000 to 21,500 Operating hours

Item	Limit	Operating hours [h]																					
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500
Engine																							
Emergency air shut-off flaps	1 w																						
Engine operation	-																						
Centrifugal oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Automatic engine oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Engine oil filter	2 a	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Emergency air shut-off flaps	3 m	X		X		X		X		X		X		X		X		X		X		X	
w = weeks m = months a = years																							

Item	Limit	Operating hours [h]																					
		11,000	11,500	12,000	12,500	13,000	13,500	14,000	14,500	15,000	15,500	16,000	16,500	17,000	17,500	18,000	18,500	19,000	19,500	20,000	20,500	21,000	21,500
Engine mounts	1 a	X		X		X		X		X		X		X		X		X		X		X	
Fuel filter	2 a	X		X		X		X		X		X		X		X		X		X		X	
Belt drive	2 a			X						X						X						X	
Air filters	3 a			X						X						X						X	
Valve gear	18 a			X						X						X						X	
Cylinder heads	18 a			X						X						X						X	
Fuel injectors	18 a			X												X							
Engine governor	18 a			X												X							
Combustion chambers	18 a			X												X							
Crankcase breathers	18 a			X																			
Engine mounts	5 a			X																			
Rubber sleeves	6 a			X																			
Component maintenance	18 a			X																			
Turbochargers	18 a			X																			
HP fuel pump	18 a			X																			
Hose lines	6 a																						
Cylinder heads	18 a																						
Extended component maintenance	18 a																						
Engine mounts	18 a																						
Torsional resilient coupling	18 a																						
Automatic engine oil filter	18 a																						
w = weeks m = months a = years																							

22,000 to 24,000 Operating hours

Item	Limit	Operating hours [h]															
		22,000	22,500	23,000	23,500	24,000											
Engine																	
Emergency air shut-off flaps	1 w																
Engine operation	-																
Centrifugal oil filter	2 a	X	X	X	X	X											
Automatic engine oil filter	2 a	X	X	X	X	X											
Engine oil filter	2 a	X	X	X	X	X											
Emergency air shut-off flaps	3 m	X		X		X											
Engine mounts	1 a	X		X		X											
Fuel filter	2 a	X		X		X											
Belt drive	2 a					X											
Air filters	3 a					X											
Valve gear	18 a					X											
Cylinder heads	18 a					X											
Fuel injectors	18 a					X											
Engine governor	18 a					X											
Combustion chambers	18 a					X											
Crankcase breathers	18 a					X											
Engine mounts	5 a					X											
w = weeks m = months a = years																	

Item	Limit	Operating hours [h]																	
		22,000	22,500	23,000	23,500	24,000													
Rubber sleeves	6 a					X													
Component maintenance	18 a					X													
Turbochargers	18 a					X													
HP fuel pump	18 a					X													
Hose lines	6 a					X													
Cylinder heads	18 a					X													
Extended component maintenance	18 a					X													
Engine mounts	18 a					X													
Torsional resilient coupling	18 a					X													
Automatic engine oil filter	18 a					X													
w = weeks m = months a = years																			

1.5 Maintenance tasks

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).	X	W1024
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.	X	W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Inspect battery-charging generator for contamination and clean if necessary.	X	W0525
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL1	500	2 a	Automatic engine oil filter	Check and clean oil indicator filter.	X	W1864
QL1	500	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).	X	W1008
QL1	1000	3 m	Emergency air shut-off flaps	Lubricate emergency air shut-off flaps at lubrication points.	X	W1247
QL1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition.	X	W1463
				Carry out visual inspection of Trunnion mounts for general condition.	X	W1575
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	3000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.	X	W1241
QL1	3000	3 a	Air filters	Fit new air filters.	X	W1005
QL1	3000	18 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul.		W1207
QL1	3000	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	6000	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	6000	18 a	Engine governor	Injector: reset drift compensation parameters (CDC).		W1713
QL1	6000	18 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	12000	18 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL3	12000	5 a	Engine mounts	Check buffer clearance of mounts, check proper seating of securing screws.	X	W1026
				Check for cracks. Measure height of rubber element.	X	W1812
QL3	12000	6 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	12000	18 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Fit new high-pressure fuel sensor.		W2004
				Check engine coolant thermostat and fit new thermal actuator.		W2006
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Overhaul engine coolant preheating system.	X	W2008
				Inspect centrifugal oil filter for wear.	X	W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
QL3	12000	18 a	Turbochargers	Fit new turbochargers.		W1041
w = weeks m = months a = years						

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
QL3	12000	18 a	HP fuel pump	Fit new high-pressure fuel pump.		W1058
QL3	24000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	24000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL4	24000	18 a	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new fuel delivery pump.		W3010
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Overhaul battery-charging generator.	X	W3042
				Replace wiring harnesses.		W3083
				Fit new rubber elements of engine mounts.	X	W3100
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	24000	18 a	Engine mounts	Fit new Trunnion mount element.	X	W1124
QL4	24000	18 a	Torsional resilient coupling	Replace rubber elements.		W1878
QL4	24000	18 a	Automatic engine oil filter	Remove filter housing and clean. Fit new filter candles.	X	W1478
w = weeks m = months a = years						